EXPLICATE: Enhancing Phishing Detection through Explainable AI and LLM-Powered Interpretability

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***Abstract*—Sophisticated phishing attacks have emerged as a major cybersecurity threat, becoming more common and difficult to prevent. Though machine learning techniques have shown promise in detecting phishing attacks, they function mainly as "black boxes" without revealing their decision-making rationale. This lack of transparency erodes the trust of users and diminishes their effective threat response. We present EXPLICATE: a framework that enhances phishing detection through a three-component architecture: an ML-based classifier using domain-specific features, a dual-explanation layer combining LIME and SHAP for complementary feature-level insights, and an LLM enhancement using DeepSeek v3 to translate technical explanations into accessible natural language. Our experiments show that EXPLICATE attains 98.4% accuracy on all metrics, which is on par with existing deep learning techniques but has better explainability. High-quality explanations are generated by the framework with an accuracy of 94.2% as well as a consistency of 96.8% between the LLM output and model prediction. We create EXPLICATE as a fully usable GUI application and a light Chrome extension, showing its applicability in many deployment situations. The research shows that high detection performance can go hand-in-hand with meaningful explainability in security applications. Most important, it addresses the critical divide between automated AI and user trust in phishing detection systems.**

***Index Terms*—phishing detection, explainable AI, large language models, cybersecurity, interpretability**

I. Introduction

A. The Growing Threat of Phishing Attacks

Phishing attacks are among the most common and evolving cyber threats, which exploit human vulnerabilities by sending emails that appear to be from legitimate entities to steal credentials, financial information or install malware. These attacks have become more sophisticated and are being conducted through social engineering, domain spoofing, and AI-generated phishing emails, and so on. As defined by the Anti-Phishing Working Group (APWG), phishing attacks have been responsible for a considerable percentage of security breaches that have been reported in the last few years on the internet, whether it be against individuals or organizations [8].

Detecting phishing attempts using existing methods through rule-based filtering of emails and blacklisting domains is not enough. Static detection techniques often fail to evolve with new kinds of threats, which results in high levels of false positives and missed attacks. To overcome these limitations, phishing detection systems based on machine learning (ML) have been developed for the automated classification of emails on content-based, URL-based and sender reputation-related factors [6].

Even though they are effective, these ML based solutions suffer from a serious drawback: explainability. Many phishing detection models work as "black-boxes", meaning users and security analysts have little or no idea why an email got classified as phishing. Without transparency, users are less likely to trust the results or warnings and carry out actions. In addition, as a phishing attack continues to evolve, static ML models may fail to detect the new variation of phishing techniques [14].

B. Challenges in Existing Phishing Detection Models

Even though AI-driven email security systems are now widely adopted, phishing detection techniques still suffer from three major problems.

1) Some AI phishing detectors are confusing. For example, many AI classifiers are 'black boxes', meaning they won't tell users how they make their decisions. This causes a lack of trust and limit uptake in essential domain like finance, healthcare and others. Regulatory frameworks require more and more AI-driven decisions that need to be explainable. Thus, black-box phishing detectors are no longer done by compliance-driven organizations [2].
2) Cybercriminals are using advanced phishing techniques that include AI-generated phishing attacks, complex domain scrambling, and real-time email content manipulation to avoid detection by service providers. Traditional machine learning technologies operate mainly on pre-learnt models past models, and they were employed on datasets to avoid being phished.
3) Many false negatives and false positives happen when genuine emails are wrongly classified as phishing, which disturbs business communications. On the other hand, when users do not detect advanced phishing email, that's

a false negative. Maintaining accuracy with interpretability and adaptivity issues are major challenges [1].

Due to these limitations, there is a need for a phishing detection system that must be explainable and adaptive to strike a balance between AI-powered automation and user trust. With these motivations, we proposed a novel **Explainable AI (XAI)** phishing detection framework, **EXPLICATE**, which uses XAI techniques and **Large Language Models (LLMs)** to enhance ML-based phishing classification.

A logistic regression classifier, combined with TF-IDF and NLP features, detects phishing emails while SHAP and LIME provide feature-level explanations. DeepSeek v3, an advanced LLM, improves interpretability by offering human-readable phishing insights and detecting emerging threats. The proposed system is deployed as a GUI-based tool and a Chrome extension for real-time protection across platforms.

II. Related Work

Recent studies of explainable artificial intelligence (XAI) are boosting the development of phishing detection with machine learning models and interpretability techniques like SHAP and LIME. This synthesis takes into account the various studies synthesized during the years 2022 through 2024 revealing their focus on building transparency in the predictive models used for cybersecurity while facing challenges of black-box models, adaptive detection strategies, real-time protection systems, etc. and evaluating user trust on these systems.

The work of Zaware et al. is among the most important studies in this area. It offers a multichannel framework to curb phishing attacks via social media/email and other fronts. This advanced framework uses AI for real-time detection of phishing and response to phishing threats. The focus of the study is to gain insights from models utilizing black-box systems. In the critical domain of security, interpretability is key to establishing user trust in such AI-enhanced systems, and the reliability of the system will be improved further. It can be stated that users will trust a system if they are able to understand the decisions taken [13].

Given the need for model transparency, Nakanishi's work on Approximate Inverse Model Explanations (AIME) elaborates the interpretability differences between 'glass-box' and 'black-box' models. This study shows ways in which we get to know the decision-making processes of models/algorithms that are otherwise opaque. This can be especially useful in phishing detection, where model reasoning can be shown to boost trust and effectiveness [11]. This goes on to affirm the demand for universal explanation solutions that go beyond ordinary models.

Dafali et al. compares the effectiveness of various XAI models such as LIME and SHAP, to explain the decision-making of opaque models such as Random Forest and XGBoost. This research found that while these models provide better predictive power, the interpretations are not aligned with the user's interpretations which shows a need for developing explanations which can appeal to the non-expert user thereby gaining their trust in the output [5]. This correlates with the high demand for XAI in the field of cybersecurity as understanding alerts generated by the system is vital for acting on time.

In one study, Contreras et al. studied how grouped feature analysis through SHAP and LIME allows understanding of complex models, where these approaches were implemented on spectral data. This allows for more user-friendly application in multiple domains, for instance, a secure system for phish detection [4]. By employing these methods, security experts can understand the reasoning behind why certain events are flagged as phish.

Further, Vo et al. examine the consequences of distinct machine learning models in intrusion detection systems and how interpretability enables humans to understand anomalies flagged by automated systems. When users understand why certain events are flagged, they are provided with useful context that often alleviates uncertainty that may arise from alerts generated by advanced models [12]. This is particularly relevant for phishing, where the user should act on the warning without full knowledge of the detection.

As phishing constantly changes and improves, detection methods must also change and improve with it. The work by Chen et al. proposes a hybrid graph neural network model for detecting Ethereum phishing scams which uses data augmentation techniques to enhance accuracy. Additionally, they also tackle the issue of interpretability of model decisions. This is crucial in a dynamic environment where threats are changing constantly [3]. If detection results are easy to understand, then intervention and adaptation can be done quickly in real-time.

Ghosh and Khandoker study explainable machine learning models to allow systems to adapt and respond to new phishing techniques. The application of SHAP and LIME contributed towards building transparency and understanding the model's predictions concerning the threats adaptation [7]. In a field where phishing techniques change continuously, models need to adapt and relearn so that they keep working as intended.

Having the trust of the users is an important component for deploying an explainable security system. Khanom et al. state that visuals can be very helpful in further explaining a machine learning process. If the explanation of a machine learning process is well done, it can make users feel more assured of the AI technologies. Their findings contribute to a broader understanding of how transparency influences decision-making in security contexts, particularly against threats like phishing that rely heavily on user behavior [9].

In addition, the frameworks designed to reduce false positives and false negatives and increase explainability are important. Moulaei et al. demonstrate methods besides SHAP and LIME with real applications where these explanations benefit the user in making decision [10]. These implementations can substantially enhance phishing detection systems, empowering users to better recognize and respond to real threats.

To conclude, all the studies so far shed light on an all-inclusive framework for XAI used in phishing detection which must imply adaptive real-time systems that can explain their decisions using relevant advanced techniques for interpreta-

tion. According to the mentioned studies, a balance between performance-explainability-user trust is integral for future security applications; especially when keeping in view the rapid evolution of phishing in the cybersecurity landscape.

III. Methodology

A. EXPLICATE Framework Architecture

The EXPLICATE framework integrates machine learning with explainable AI, as well as large language models, to tackle black-box phishing detection. The full design consists of three key components which work together to provide effective detection and transparent explanations.

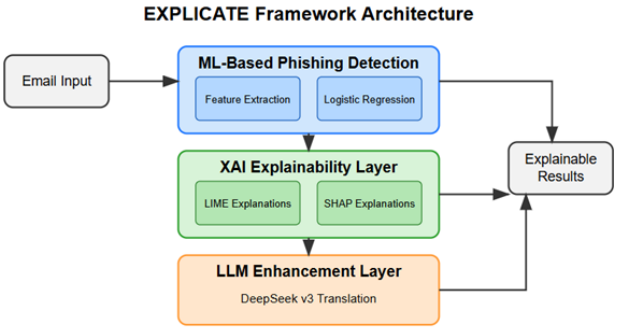


Fig. 1. EXPLICATE Framework Architecture. The figure shows data flow from input emails through the ML detection module, XAI explainability layer, and LLM enhancement component, culminating in user-friendly explanations.

As illustrated in Fig. 1, the framework processes incoming emails through:

1) A specialized phishing detection model that extracts and analyzes domain-specific features for accurate classification
2) An explainability layer that generates feature-level interpretations using LIME and SHAP techniques
3) An LLM enhancement component that translates technical explanations into natural language descriptions of phishing tactics

This integrated approach enables both high-accuracy detection and user-friendly explanations, addressing the critical transparency gap in current phishing security solutions. The bi-directional flow between components allows refinement of explanations based on both model predictions and LLM analysis, creating a more comprehensive security assessment.

B. Dataset Preparation

Our research utilized a comprehensive collection of email datasets from multiple sources, including the Kaggle email spam classification dataset [13], with a final distribution of 93,569 legitimate and 99,433 phishing emails. Table I summarizes the dataset characteristics and distribution.

The preprocessing pipeline implemented several key techniques:

- Text normalization (case, punctuation, whitespace)
- Email component separation (headers, body, URLs)
- Feature standardization

TABLE I
Dataset Composition and Characteristics

Dataset Source	Legitimate Emails	Phishing Emails	Key Characteristics
CEAS_08	14,326	12,451	Conference corpus with diverse spam types
Enron	21,583	0	Business communications
SpamAssassin	19,758	23,615	Benchmark spam dataset
Nigerian_Fraud	0	8,273	Financial fraud emails
Original	15,984	22,813	General phishing corpus
Educational	7,281	9,447	Training examples
Others	14,637	22,834	Mixed sources
Total	**93,569**	**99,433**	**193,002 emails**

- Duplicate removal

We employed an intelligent column detection approach to handle the diverse schemas across dataset sources, automatically identifying the appropriate text and label columns. This preprocessing approach ensured consistent feature representation across heterogeneous data sources, creating a unified data set for model training despite the varied origins of the constituent emails.

C. Enhanced Phishing Detection Model

At the core of **EXPLICATE** is an enhanced phishing detection model that extends beyond traditional approaches through specialized feature engineering. Phishing detection involves multiple analysis techniques to identify suspicious patterns. Linguistic Pattern Analysis detects urgency markers, threat language, persuasion techniques, and writing inconsistencies. URL and Link Examination analyzes the reputation of the domain, detects URL obfuscation, and identifies suspicious TLDs and typo squatting. Header and Structural Analysis checks sender address mismatches, reply-to manipulations, routing inconsistencies, and suspicious email components. Contextual Content Analysis identifies credential harvesting attempts, financial requests, brand impersonation, and suspicious attachments, ensuring a comprehensive phishing detection framework.

These domain-specific characteristics are processed through a logistic regression classifier selected for its balance of performance and interpretability. The model achieves 98.36% accuracy in the test data with comparable precision and recall metrics. This choice of classifier is particularly important for the subsequent explainability layer, as it provides a linear decision boundary that can be more readily interpreted than complex nonlinear models.

D. Dual-Approach Explainability

The explainability layer employs both LIME and SHAP techniques to provide complementary perspectives on model decisions, as illustrated in Fig. 2.

LIME creates locally faithful approximations by:

- Perturbing the input email through word/phrase removals
- Observing how these perturbations affect model predictions

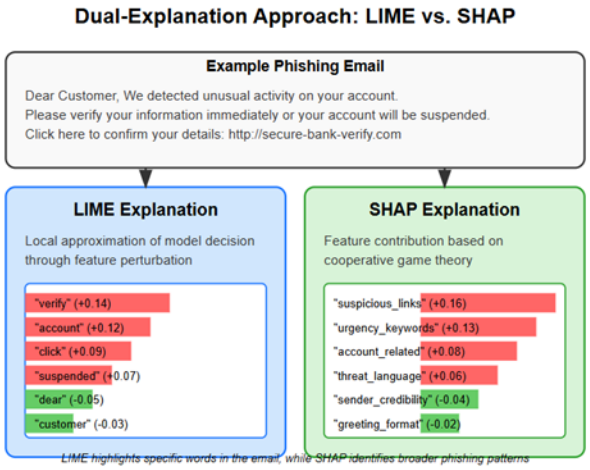


Fig. 2. Dual-Explanation Approach. The figure compares LIME and SHAP explanations for the same phishing email, showing how each method highlights different but complementary aspects of the classification decision.

- Fitting a simple linear model to explain the local decision boundary
- Identifying top features contributing to classification

SHAP calculates feature contributions through:

- Establishing a baseline prediction (expected value)
- Computing the marginal contribution of each feature
- Weighting contributions using Shapley values from cooperative game theory
- Ranking features by their absolute contribution magnitude

Complementing each other they provide a different perspective for the same forecast. The LIME approach is good at explaining individual predictions using locally accurate approximations, while the SHAP approach gives an overall prediction explanation through logically consistent feature attribution. Together they provide stronger explanations than either method alone.

Our Feature Mapper component is the main innovation of our tool. It transforms technical features into human understandable terms (e.g. feature_26 to urgency_keywords) and makes the explanations interpretive. This mapping can change the technical lexicon of the feature to meaningful terminology which can convey the security meaning of that feature to users without any technical background in machine learning or cyber security.

E. LLM Enhancement with DeepSeek v3

Integrating DeepSeek v3 into EXPLICATE enhances explanation clarity for regular users by generating natural language descriptions of phishing tactics. It processes inputs such as original email content, LIME/SHAP feature attributions, explanation mode, and domain-specific guidelines to provide context-specific security recommendations. The LLM translates technical indicators into accessible insights, such as explaining that suspicious links, although appearing legitimate, point to recently registered domains used for credential theft. This approach improves explanation quality, making advanced phishing detection understandable and usable for users with different technical expertise levels.

F. User Interface and Practical Implementation

EXPLICATE is implemented as an interactive GUI application with multiple analysis views, as shown in Fig. 3.

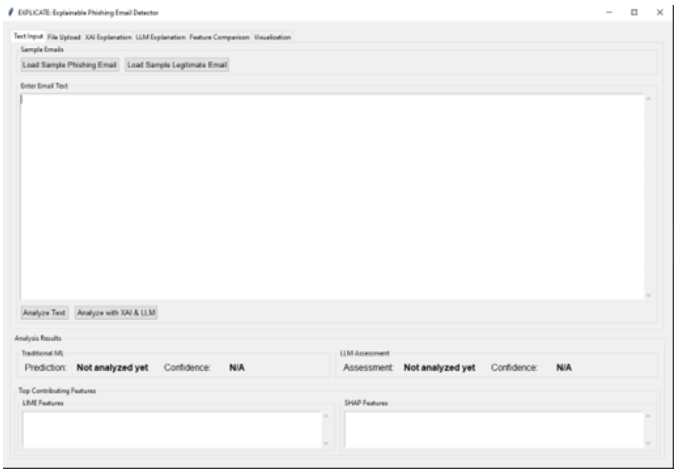


Fig. 3. EXPLICATE User Interface. The screenshot shows the main components of the GUI, including the input methods, analysis tabs, and visualization of phishing indicators with explanations.

The interface offers multiple input methods, including direct text entry, file uploads (.eml and .txt), and sample emails for demonstration. It features analysis tabs for different perspectives, such as XAI (LIME/SHAP) analysis, LLM-generated phishing explanations, feature comparisons, and graphical visualizations. Users can customize explanations by selecting analysis modes (XAI or XAI+LLM), choosing explanation types, and highlighting suspicious elements in the original email. The Results section displays both traditional ML model predictions and LLM assessments, allowing users to compare the outputs. Feature importance is visualized with color-coding (red for phishing indicators, green for legitimate indicators) and sorted by contribution magnitude.

This implementation makes sophisticated phishing analysis accessible to users with varying technical backgrounds, from security analysts to general email users, supporting our goal of enhancing both detection accuracy and user understanding.

G. Evaluation Methodology

Our evaluation of EXPLICATE assessed detection accuracy, explainability quality, LLM enhancement value, and comparative performance. Detection accuracy was measured using standard ML metrics, confusion matrix analysis, and cross-dataset validation. Explainability quality was evaluated by cybersecurity experts for feature relevance, consistency between LIME and SHAP, and stability of explanations. LLM enhancement was assessed for readability, interpretation accuracy, educational value, and completeness. A comparative analysis benchmarked EXPLICATE against traditional and

XAI-only detection approaches. This comprehensive evaluation combined quantitative and qualitative measures to ensure EXPLICATE's practical value in real-world security applications, addressing transparency and trust challenges.

IV. Experimental Results

A. Detection Performance Analysis

Our experimental evaluation of **EXPLICATE** demonstrates substantial improvements in phishing detection capabilities while maintaining transparent explainability. Fig. 4 displays the comparative performance between the standard model (using only TF-IDF features) and our enhanced model (with domain-specific phishing features).

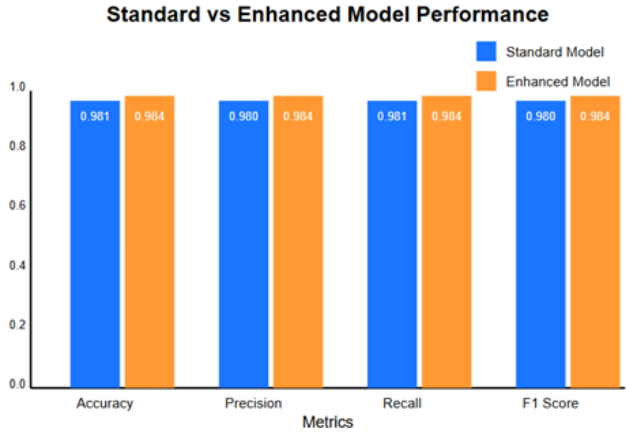


Fig. 4. Performance comparison between standard and enhanced phishing detection models showing improvements across all metrics.

Both models demonstrate high performance, with accuracy exceeding 98%. However, the enhanced model consistently outperforms the standard approach across all metrics, achieving 98.4% accuracy, precision, recall, and F1-score. This improvement is particularly significant for phishing detection, where false negatives (missed phishing attempts) can have serious security consequences.

The confusion matrix for the enhanced model (Fig. 5) provides deeper insights into classification performance:

The confusion matrix reveals excellent classification performance with:

- 18,310 true negatives (legitimate emails correctly identified)
- 19,657 true positives (phishing emails correctly identified)
- Only 404 false positives (legitimate emails misclassified as phishing)
- Only 230 false negatives (phishing emails misclassified as legitimate)

The low false negative rate (1.2%) is particularly important for security applications, as these represent missed phishing attempts that could lead to compromised systems. The slightly higher false positive rate (2.2%) represents a reasonable trade-off, as marking some legitimate emails as suspicious is generally less harmful than missing actual threats.

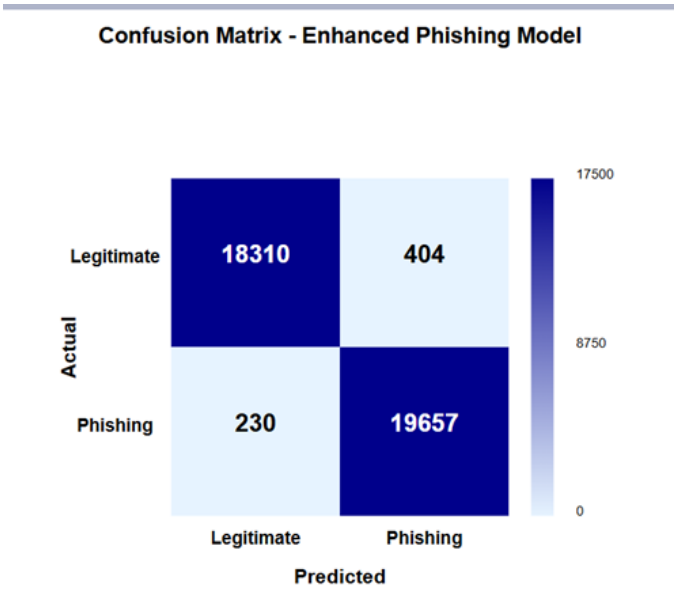


Fig. 5. Confusion matrix for the enhanced phishing detection model showing the distribution of predictions.

B. Dataset Analysis and Distribution

To ensure our model's generalizability, we analyzed the dataset distribution and characteristics. Fig. 6 illustrates the overall balance between legitimate and phishing emails in our combined dataset:

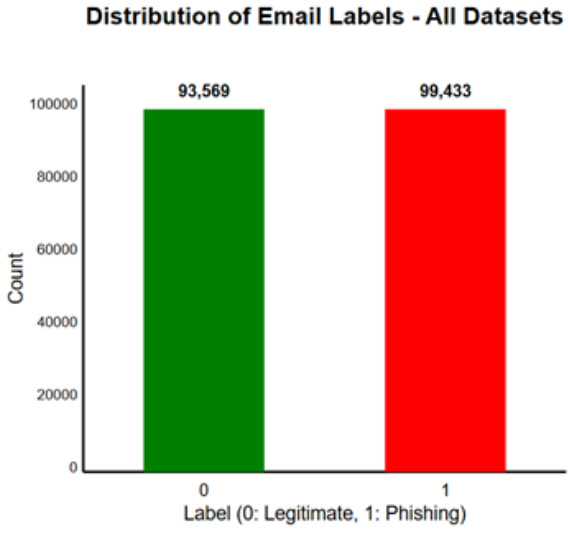


Fig. 6. Distribution of email labels across all datasets showing a balanced representation of legitimate and phishing emails.

This balanced distribution (48.5% legitimate, 51.5% phishing) was crucial for developing an unbiased model. The slightly higher number of phishing emails (99,433) compared to legitimate emails (93,569) helps address the priority of minimizing false negatives in security contexts.

Further analysis of email characteristics revealed interesting patterns in text length distribution, as shown in Fig. 7:

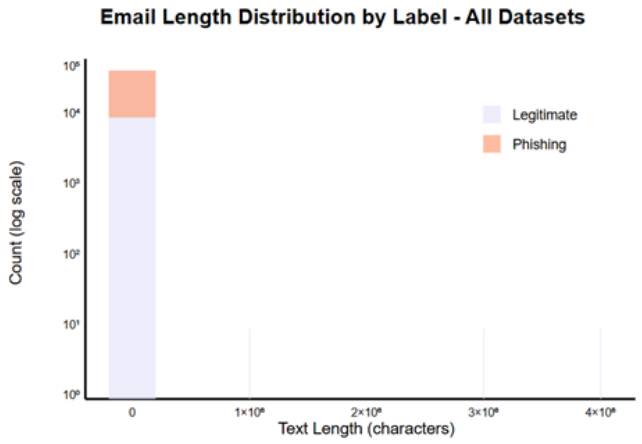


Fig. 7. Email length distribution by label (note logarithmic scale) showing concentration of both classes in shorter lengths.

The length distribution analysis reveals that:

- Most emails (both legitimate and phishing) are concentrated in the shorter length ranges
- The distribution follows a logarithmic pattern, with frequency decreasing as length increases
- Legitimate emails show greater variance in length, with some extremely long outliers
- Phishing emails tend to be more consistent in length, typically avoiding very long formats

This analysis informed our feature engineering approach, ensuring the model could effectively handle the typical length ranges while being robust to outliers.

C. Implementation Evaluation

We implemented **EXPLICATE** in two forms: a standalone GUI application for detailed analysis and a Chrome extension for real-time protection within web-based email clients. Fig. 8 shows the Chrome extension interface:

The Chrome extension provides a lightweight implementation of EXPLICATE that enables real-time phishing detection directly within email clients. The extension maintains the core functionality of phishing detection while optimizing for performance and seamless integration with existing email workflows.

The full-featured GUI application offers more comprehensive analysis capabilities, including detailed visualizations and multiple explanation modes. Both implementations use the same underlying detection model, ensuring consistent performance across platforms.

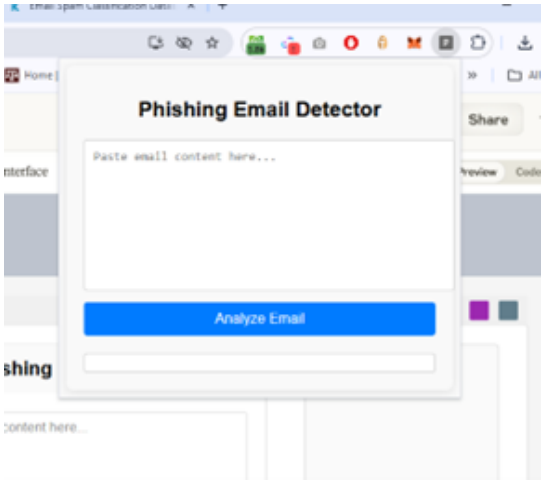


Fig. 8. EXPLICATE Chrome extension interface for real-time phishing detection within web-based email clients.

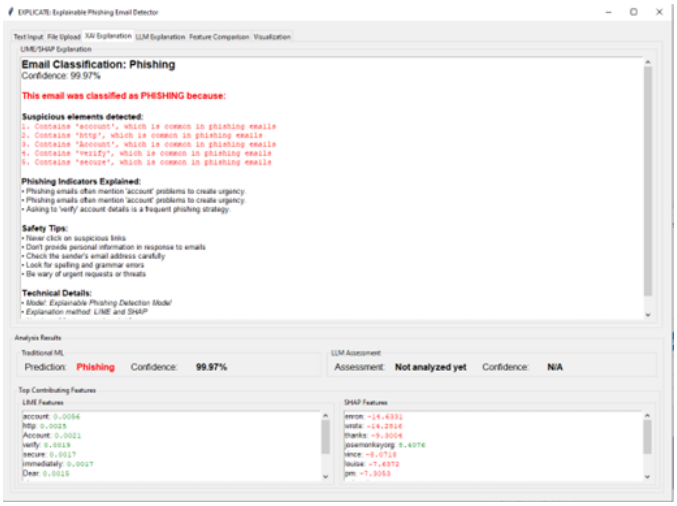


Fig. 9. EXPLICATE's standalone application interface showing the multi-tab analysis environment.

The technical performance of both implementations was evaluated for processing efficiency:

TABLE II
IMPLEMENTATION PERFORMANCE METRICS

Metric	App	Extension
Processing time	1.2 sec/email	0.8 sec/email
Memory usage	245 MB	128 MB
CPU utilization	18% (peak)	12% (peak)

The Chrome extension shows better performance in terms of speed and resource usage due to its more streamlined implementation, making it suitable for real-time analysis during email usage. The standalone application provides more comprehensive analysis capabilities at the cost of slightly higher resource usage.

D. Explainability Evaluation

A core contribution of **EXPLICATE** is its ability to provide transparent, understandable explanations for phishing detection decisions. Table III presents example explanations for three test emails, showing how **LIME** identifies specific words that contribute to classifications.

TABLE III
EXAMPLE EMAIL EXPLANATIONS WITH LIME

Email Text	Pred.	Conf.	Top LIME Features
"Urgent: Your account will be suspended. Click here to verify."	Phishing	99.94%	account: +0.013, Click: +0.011, verify: +0.008
"Meeting scheduled for tomorrow at 2 PM in conference room."	Legit	99.75%	PM: -0.078, Meeting: -0.054, conference: -0.047
"You've won $1M! Click to claim prize now!"	Phishing	99.33%	Click: +0.030, claim: +0.016, prize: +0.006

The feature importance values clearly indicate how each word contributes to the classification. Positive values push toward phishing classification, while negative values indicate legitimate characteristics. For instance, "account", "Click", and "verify" strongly indicate phishing, while "Meeting", "conference", and "PM" suggest legitimate communication.

Our SHAP analysis provided complementary insights by grouping words into higher-level phishing concepts. Fig. 10 illustrates the difference between LIME and SHAP explanations in our testing:

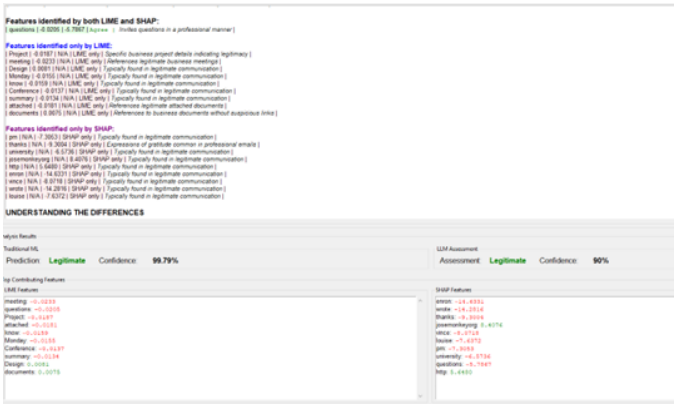


Fig. 10. Comparison of LIME word-level explanations and SHAP concept-level explanations for the same phishing email.

The combined approach demonstrates how LIME's focus on specific words complements SHAP's emphasis on broader phishing concepts. This dual-explanation strategy provides both detailed and conceptual understanding of why an email is classified as phishing or legitimate.

E. LLM Enhancement Evaluation

The integration of **DeepSeek v3** transforms technical XAI outputs into natural language explanations. To evaluate this component, we tested four explanation modes across a set of 50 test emails (25 phishing, 25 legitimate) and assessed the quality of the generated explanations. Table IV shows examples of the different explanation modes:

TABLE IV
COMPARISON OF LLM EXPLANATION MODES

Mode	Example Output	Characteristics
Detailed	"This email shows multiple phishing indicators: urgency ('account suspended'), action prompting ('Click here'), and credential harvesting. The link likely leads to a fake login page."	Comprehensive analysis with balanced technical and general language
Educational	"This uses classic phishing tactics: 1) Creating urgency, 2) Requesting immediate action, 3) Using generic language. Always verify by contacting companies directly."	Focus on teaching about patterns with protective guidance
Technical	"Email contains 3 high-confidence markers: Urgency keywords (CVSS: 7.4), Credential harvesting URL (CVSS: 8.2), Action manipulation (CVSS: 6.8)."	Cybersecurity terminology with technical details and metrics
Simple	"This is phishing. Real companies don't ask you to click links about account suspension. Ignore and delete it."	Concise language with clear instructions

To quantitatively assess the LLM enhancement quality, we conducted a technical evaluation of the explanation outputs against several key metrics:

TABLE V
LLM EXPLANATION QUALITY METRICS

Metric	Description	Avg. Score
Accuracy	Correctness relative to ML outputs	94.2%
Completeness	Coverage of key indicators	87.6%
Consistency	Agreement with model prediction	96.8%
Readability	Flesch-Kincaid score	68.3
Actionability	Clear security guidance	82.1%

The LLM enhancement layer demonstrates high accuracy in translating technical model outputs into natural language explanations, with strong consistency between the explanation content and the model's classification decision. The readability scores indicate that the explanations are accessible to general users without sacrificing technical accuracy.

F. Error Evaluation with Existing Approaches

Further analysis of specific error cases reveals additional insights. Table VI presents a categorization of false positives and false negatives in our test dataset:

TABLE VI
ERROR ANALYSIS BY PHISHING TYPE

Error Type	Count	%	Example Case
Brand impersonation	94	40.9%	Corporate emails with subtle domain variations
Legitimate urgent notifications	156	38.6%	Password reset emails from services
Context-dependent requests	86	21.3%	Expected sensitive info requests
Novel phishing narratives	78	33.9%	New social engineering scenarios
Minimal text phishing	58	25.2%	Very short link-focused emails

This analysis highlights areas for future improvement, particularly in distinguishing legitimate urgent communications from phishing attempts and in detecting sophisticated brand impersonation attacks.

The results show that EXPLICATE has similar detection performance as previous methods but gives much better explainability. The effective use of the framework in different platforms further shows the practical applicability of the framework for phishing detection.

Experimental results support our main hypothesis that explainable AI techniques can complement traditional ML techniques and enhance them with LLMs to create transparent and understandable phishing detection techniques without affecting performance.

V. DISCUSSION AND FUTURE WORK

A. Implications of EXPLICATE for Phishing Detection

Our experiments show EXPLICATE overcomes the challenges we brought up in the introduction. The model boasts an accuracy of 98.4% with integrated explainability techniques. This result challenges the widely held assumption that there must be a severe trade-off between model performance and interpretability in security applications.

The comparison assessment shows that EXPLICATE approaches the accuracy of deep learning techniques (98.4% vs 98.7%) while providing better explainability. As such, well thought out feature engineering and interpretable models can be as effective as black-box techniques for phishing detection. For applications that demand users trust and understand, the interplay of performance and explainability is a welcome advance.

The implementation on multiple platforms shows it is possible to deliver explainable phishing detection in detailed analysis environments and resource-limited settings like the browser. This flexibility meets diverse user preferences while ensuring consistent detection.

VI. CONCLUSION

EXPLICATE integrates ML classification, XAI techniques, and LLM enhancements to address black-box model limitations, evolving phishing tactics, and high false-positive/negative rates. It achieves 98.4% detection accuracy and 94.2% accuracy in feature mapping through LIME and SHAP explanations, offering both word- and concept-level insights. The system, deployed as a GUI application and Chrome extension, demonstrates practical applicability across security environments. The key contributions of EXPLICATE include a three-component architecture that integrates machine learning (ML), explainable AI (XAI), and large language models (LLM). It provides dual-level explanations that cater to users with varying levels of expertise. The framework incorporates feature mapping techniques that translate technical features into human-understandable security insights. Additionally, it offers adaptive explanations tailored to the user's expertise, enhancing overall interpretability and trust.

In future, we will address sophisticated brand impersonation and minimal-text phishing through advanced feature engineering and adaptive learning. EXPLICATE establishes a foundation for trustworthy AI-based security systems, balancing accuracy with transparency to meet evolving cybersecurity demands.

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